

Name: _____

Date: _____

RATIO AND PROPORTION, (DIVIDING QUANTITIES IN A GIVEN RATIO)

LO: I can divide quantities in a given ratio and solve related problems.

Dividing quantities

To **divide a quantity** in a ratio, add the parts to find the **total number of shares**. Divide the quantity by this total to find **one share**.

Then: Multiply each ratio number by the value of one share to find each portion.

Quick examples

●	£120 in 3:5 shares = 8, one share = 15	$120 \div 8 = 15$
●	Result: £45 and £75	3×15 and 5×15
●	Check: $45 + 75 = 120$	always verify
●	£120 in 3:5 gives £40 each	not equal sharing
●	Total shares for 3:5 is 15	total shares = $3 + 5 = 8$

Key idea

Total shares = sum of ratio. One share = quantity $\div$ total shares. Each portion = ratio number $\times$ one share.

**Quantity $\div (a + b) = \text{one share};$
portions = $a \times \text{share}, b \times \text{share}$**

Always check the portions add to the original

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – basic division

Divide 200 ml in the ratio 3:7.

$$\text{Total shares} = 3 + 7 = 10$$

$$\text{One share} = 200 \div 10 = 20 \text{ ml}$$

$$3 \text{ shares} = 60 \text{ ml}, 7 \text{ shares} = 140 \text{ ml}$$

60 ml and 140 ml

$$\text{Check: } 60 + 140 = 200 \text{ ml.}$$

Example 2 – money sharing

£240 is shared between Amy and Bella in ratio 5:3. How much does each get?

$$\text{Total shares} = 5 + 3 = 8$$

$$\text{One share} = 240 \div 8 = \text{£}30$$

$$\text{Amy: } 5 \times 30 = \text{£}150, \text{ Bella: } 3 \times 30 = \text{£}90$$

Amy £150, Bella £90

$$\text{Check: } 150 + 90 = 240.$$

Example 3 – finding total from one part

Jo and Kim share sweets in ratio 2:5. Kim gets 35 sweets. How many in total?

$$\text{Kim's 5 shares} = 35$$

$$\text{One share} = 35 \div 5 = 7$$

$$\text{Total} = (2 + 5) \times 7 = 49$$

49 sweets total

Work backwards: divide the known amount by its share number.

Example 4 – three-way split

Split 450 g into the ratio 2:3:4.

$$\text{Total shares} = 2 + 3 + 4 = 9$$

$$\text{One share} = 450 \div 9 = 50 \text{ g}$$

$$\text{Portions: } 100 \text{ g, } 150 \text{ g, } 200 \text{ g}$$

100 g, 150 g, 200 g

$$\text{Check: } 100 + 150 + 200 = 450.$$

YOUR TURN – PRACTICE (deliberate practice)

1. Two-part divisions

Divide each quantity in the given ratio.

- a) £90 in 1:2
- b) 180 cm in 4:5
- c) £350 in 3:4
- d) 1 kg in 2:3

2. Three-part divisions

Divide each quantity in the given ratio.

- a) £180 in 1:2:3
- b) 400 ml in 2:3:5
- c) £600 in 3:4:5
- d) 72 in 1:3:5

3. Working backwards

Find the total or the other part.

- a) Ratio 2:3. Smaller part = 18. Total?
- b) Ratio 4:5. Larger part = 60. Smaller?
- c) Ratio 1:6. Smaller = 9. Total?
- d) Ratio 3:7. Difference = 20. Total?

4. Word problems

Solve each ratio problem.

- a) Cordial:water = 1:5. Make 900 ml. Cordial needed?
- b) Boys:Girls = 5:4 in a class of 36.
- c) £1500 inheritance split 2:3. Larger share?
- d) Paint: red:yellow 3:1. Need 2 litres total. Red?

5. Harder problems

Solve each problem.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Total shares for ratio 3:5 = 8

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Correct answer: _____

Reason: _____

b) Dividing £100 in ratio 1:4 gives £25 and £75

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Correct answer: _____

Reason: _____

c) One share = total ÷ sum of ratio parts

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Correct answer: _____

Reason: _____

d) Ratio 2:3 means 2 out of 3

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Correct answer: _____

Reason: _____

e) The portions must add up to the original total

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Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Three charities share a **£12,000** donation in the ratio **2:3:7**. The largest charity then donates **20%** of its share to the smallest. How much does each charity have now?

Expression: _____

Simplified: _____

b) A rectangular garden has length to width in ratio **4:3**. The perimeter is **56 m**. Find the dimensions and the area of the garden.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

